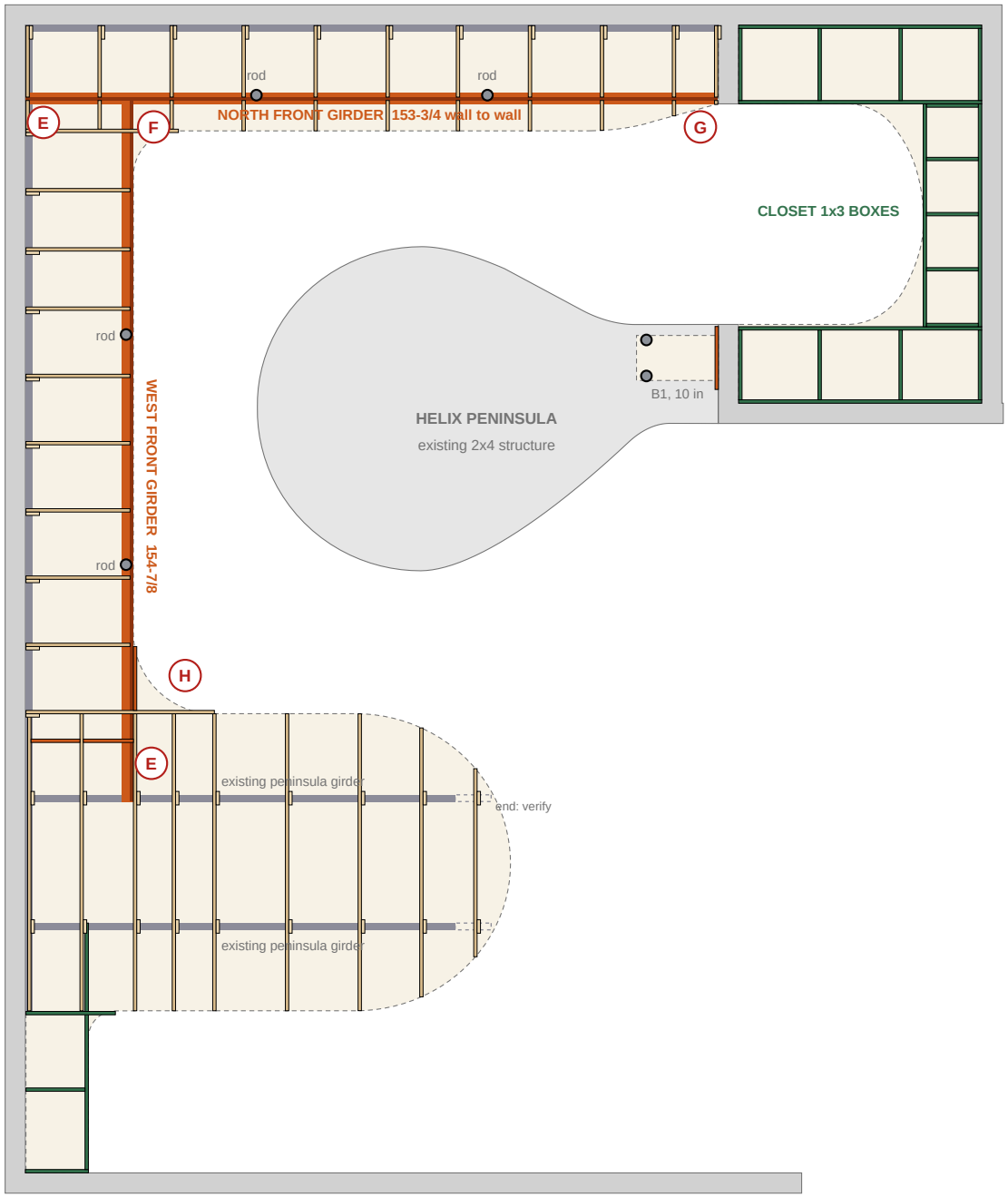
Framing plan - main level, deck removed

Looking down. Orange = new front girders; grey = existing girders.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



LEGEND

- New front girder: 1x2 web (dark) on a 1x3 flange
- New 1x2 joist or lookout
- 1x3 riser block on a -3.25 girder
- 1x2 blocking under a deck seam
- Closet 1x3 boxes / grid (wall side, rail, cross piece)
- Existing L-girder flange
- 3/8 rod, 2 per run, over a staging leg

NOTES

Joists run north-south on the north shelf, east-west on the west shelf and north-south on the SW peninsula, about 16 in o.c. Every deck seam lands on a joist or on blocking.

North girder set back to the wall-stub end, web centred on its flange: joists butt its back, 1x2 lookouts in line carry the deck to the front edge (up to 7 in). West girder web on the deck edge.

Rods: nominal at the third points (north x 56, 107; west y 116, 167). Move each one over a staging leg; keep every gap under 60 in.

E: front girder seats on an existing girder (sheet 4). F: west girder into the north girder. G: north girder at the wall stub. H: rim and blocking under F3.

Peninsula girders and joist ends at the tip: drawn to x 100; check the as-built girders before cutting the last two joists.

Closet past the helix: three 1x3 boxes on edge. North and south run full depth to the east wall; a short 13 in box fills the east wall between them, its front rail under the inside edge of the helix track and a cross piece under the B2/B3 seam (y 89.5). Cross pieces 18 in o.c. in the side boxes, 12-1/4 in the east box.

B1 (0 in board, helix to the south opening): on the helix at its post and on a 1x3 joist on the wall stub. Past the curve it is 10 in wide to the wall: two 3/8 rods 4 in off the centre line, halfway; a temporary post under it until then.

JOIST STATIONS (centres, room inches)

North, x: 5, 21, 37, 53, 69, 85, 100-5/8, 116-5/8, 132-5/8, 148-5/8, 158

West, y: 71, 84-1/8, 97-3/8, 110-1/2, 125-1/2, 140-3/8, 155-1/4, 170-1/4, 185-1/4, 200-1/8

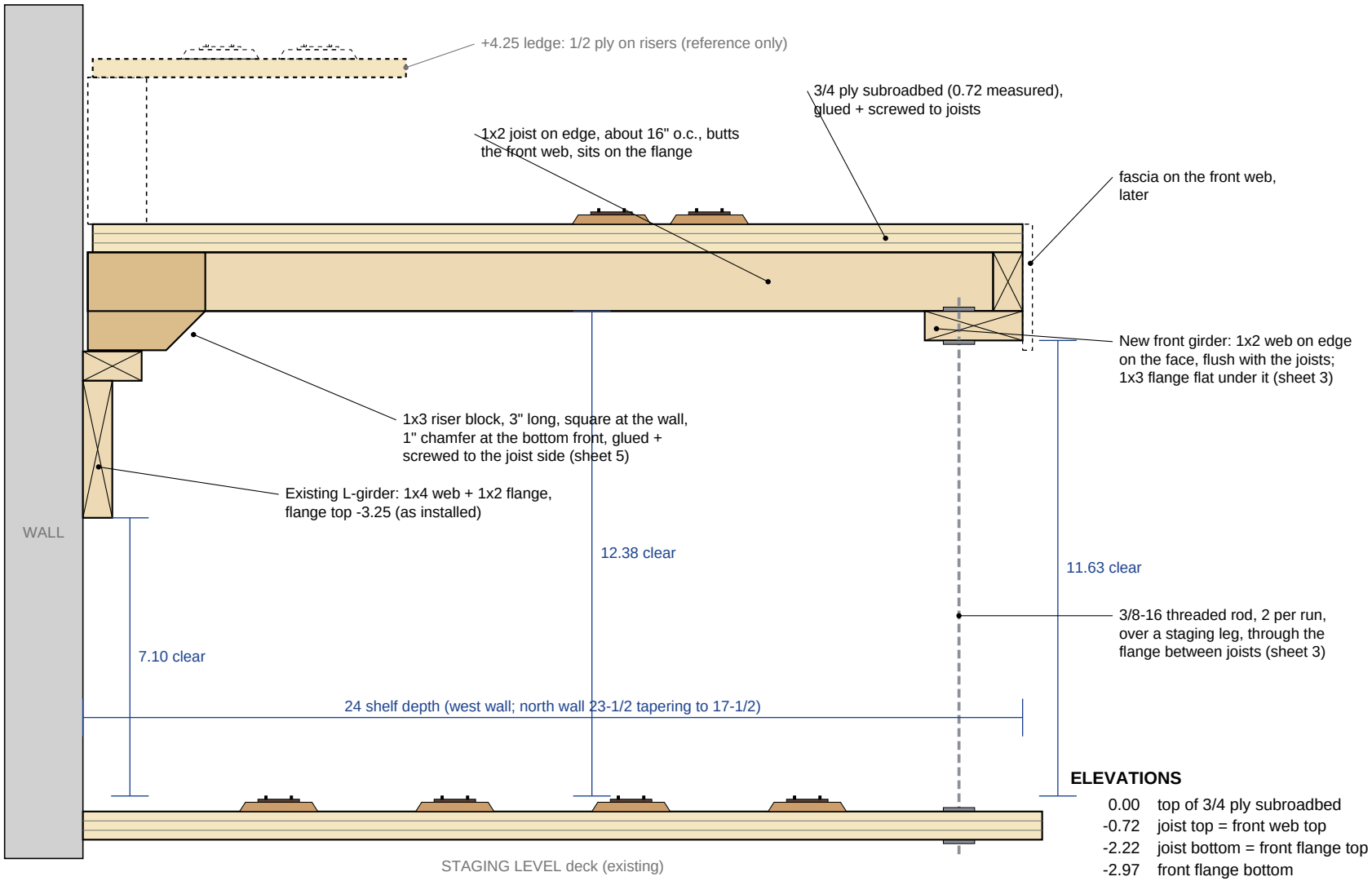
Peninsula, x: 5-3/8, 17, 28-7/8, 37-1/2, 46-1/2, 62-5/8, 78-3/4, 92-1/2, 104-1/2

Room inches: x east from the west wall corner, y south from the north wall line (same as the cut plan). Plan about 1:39 - work from the stations, not the drawing.

Typical shelf section - west wall (north wall the same, girder set back)

Section through a joist, perpendicular to the wall.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



ELEVATIONS

0.00	top of 3/4 ply subroadbed
-0.72	joist top = front web top
-2.22	joist bottom = front flange top
-2.97	front flange bottom
-3.22	riser block bottom
-3.25	back girder flange top
-7.50	back girder web bottom
-14.60	staging railhead
-15.00	staging deck top

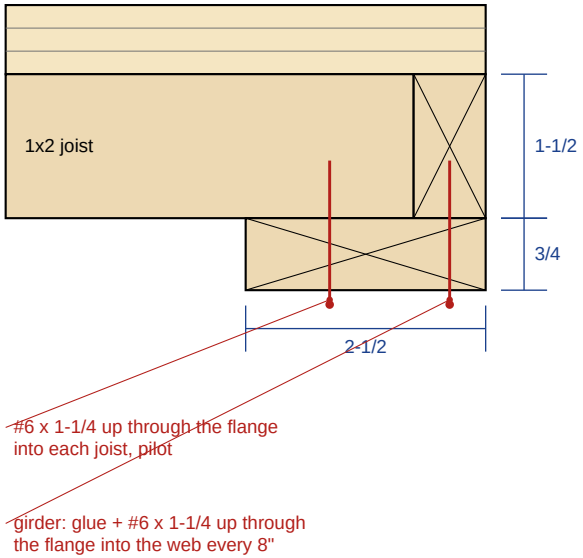
Block bottom lands 0.03 above the back flange (3.25 - 2.50 - 0.72); the deck top finishes 1/32 low - within tolerance, or shim the flange. The joists butt the front web and sit on its flange - no blocks at the front.
North wall: the girder is set back to the wall-stub end with its web centred; 1x2 lookouts in line with the joists carry the deck to the front edge (sheet 3, N). Track = 0.236 cork + 0.160 Atlas code 83.
Scale 1:4 on Letter landscape. Lumber cut in section shows an X; the joist and riser block are in side view.

Details - front girder: 1x2 web on the face, 1x3 flange under it

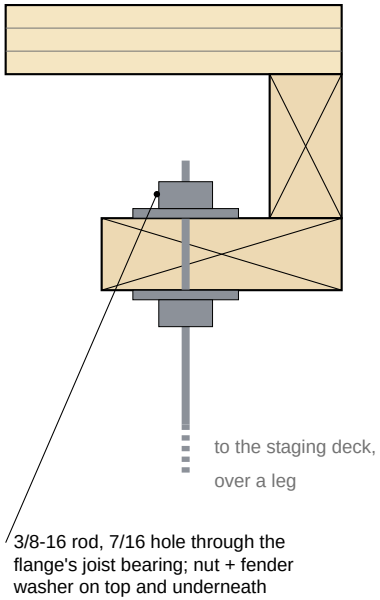
C: west, at a joist. D: between joists, at a rod. N: north girder, joist and lookout. All cut across the girder.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

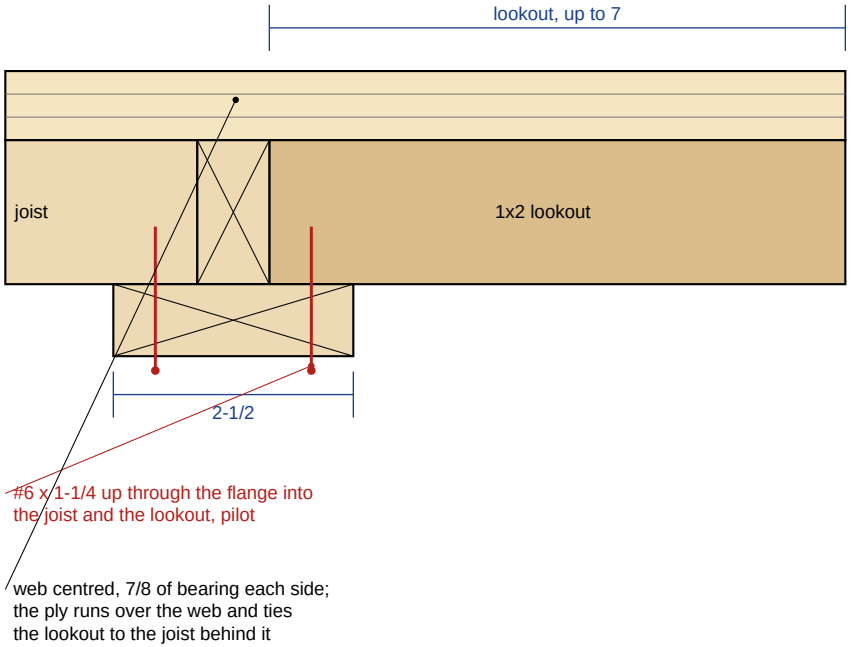
C WEST, AT A JOIST 1:2



D BETWEEN JOISTS, AT A ROD 1:2



N NORTH, JOIST AND LOOKOUT 1:2



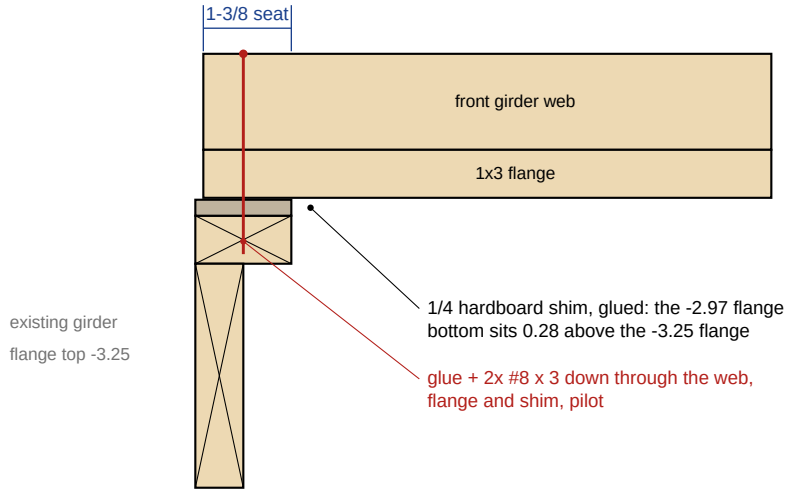
Build each girder on the bench: lay the 1x3 flange flat, glue the 1x2 web on edge on top - on the outer edge for the west girder, centred for the north - and screw up through the flange every 8". One 16 ft piece of each per run. The web top is the joist top (-0.72), so the ply lies flush across both. Joists butt the web and sit on the flange. Lookouts on the north run sit in line with each joist. Drill the rod holes between joists, over a staging leg, two per run.

Details - girder junctions

E: seat on an existing girder. F: west girder into the north girder. G: wall stub. H: under F3.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

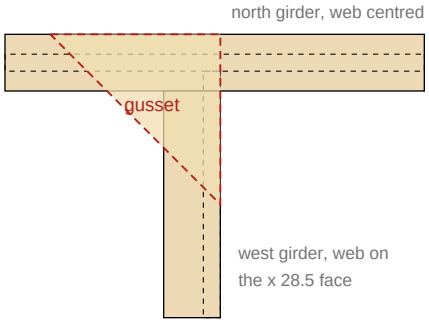
E SEAT ON AN EXISTING GIRDER 1:3



At the west wall: the north girder on the west wall girder, end 1/8 off the wall.

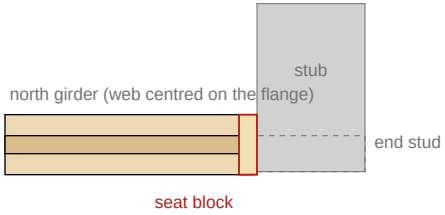
At the peninsula: the west girder across the peninsula's north girder flange.

F WEST GIRDER INTO THE NORTH GIRDER, FROM BELOW 1:8



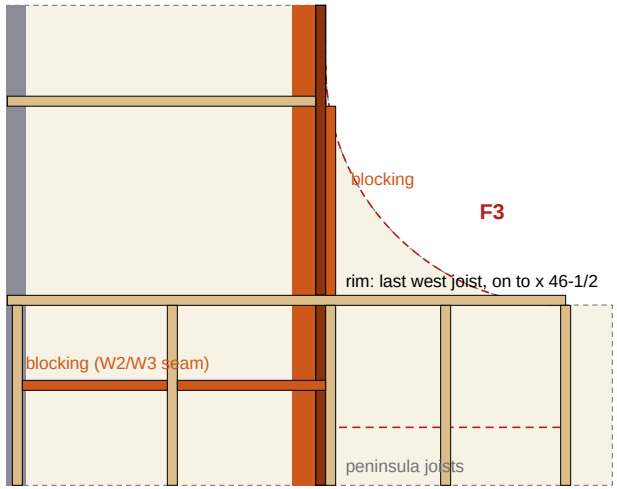
1/2 ply gusset, 7-1/2 x 7-1/2 triangle, under both flanges: glue + 3x #6 x 1-1/4 up into each flange, pilot. The west flange butts the north flange's front edge; the west web runs on over the north flange to the north web: glue, 2x #6 x 2 through the north web into its end. Webs dashed.

G NORTH GIRDER AT THE WALL STUB, PLAN 1:8



Seat block: 1x3 x 4 long, flat on the wall at the end stud, top at -2.97 under the flange end. 2x #8 x 2-1/2 through the block into the end stud; #8 x 3 down through the web and flange into the block. Girder end 1/8 off the wall. If there's no stud at the end of the stub, add blocking in the wall first.

H RIM AND BLOCKING UNDER F3, PLAN 1:14



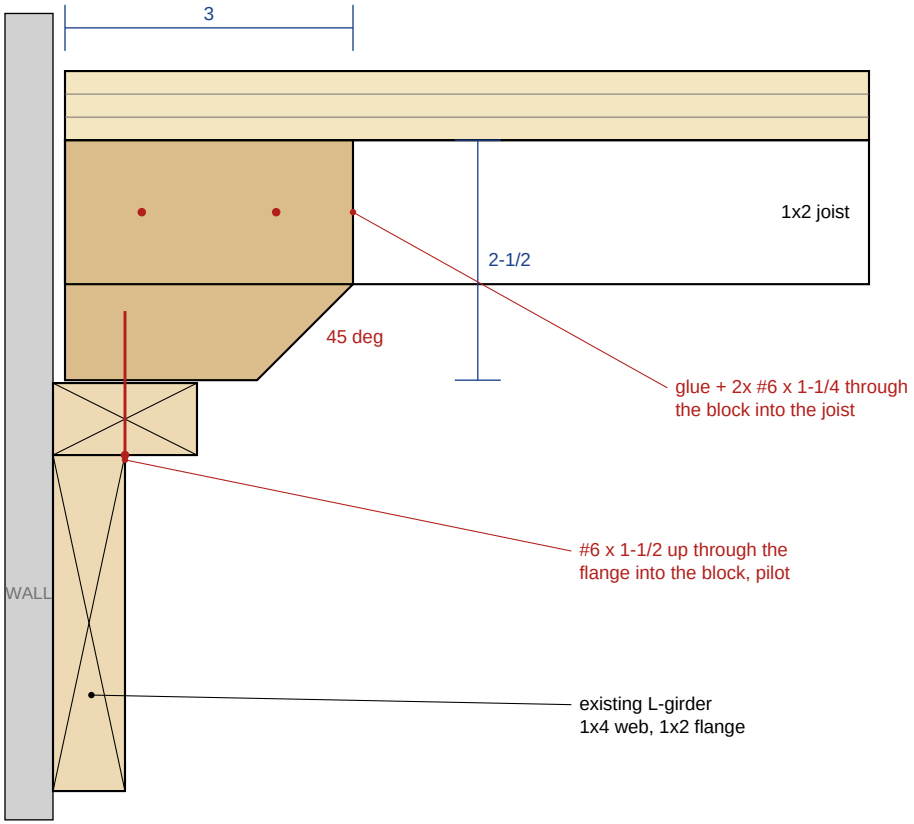
The rim crosses the west girder on its flange; the web is cut either side, 2x #6 x 2 through the rim into each web end. Peninsula joists butt the rim: 2x #6 x 2 each. The NW joist (y 71) crosses the same way. Blocking: 1x2 on edge between joists, ends screwed 2x #6 x 2.

Details - riser block at a -3.25 girder (back wall and peninsula)

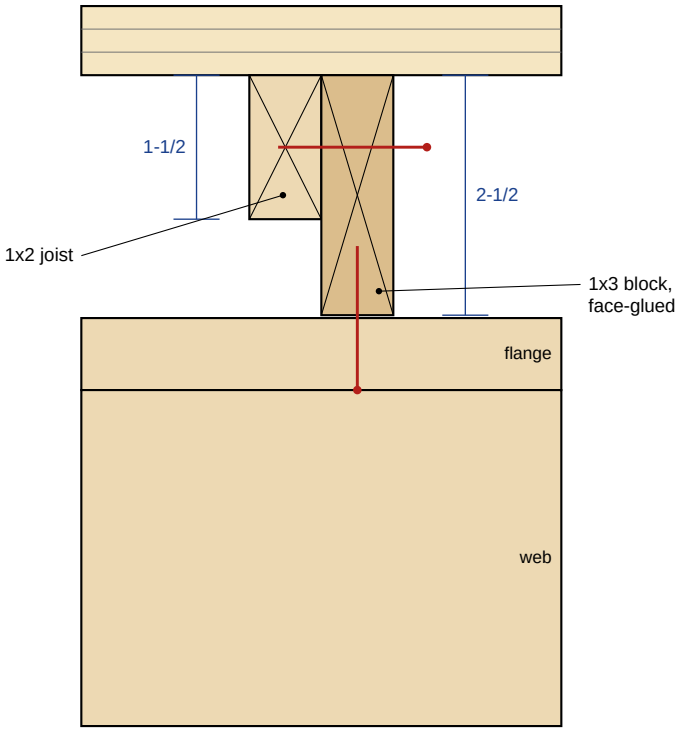
A: side view along the joist. B: looking along the joist (joist and block cut, girder in elevation).

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

A SIDE VIEW ALONG THE JOIST 1:2



B LOOKING ALONG THE JOIST 1:2

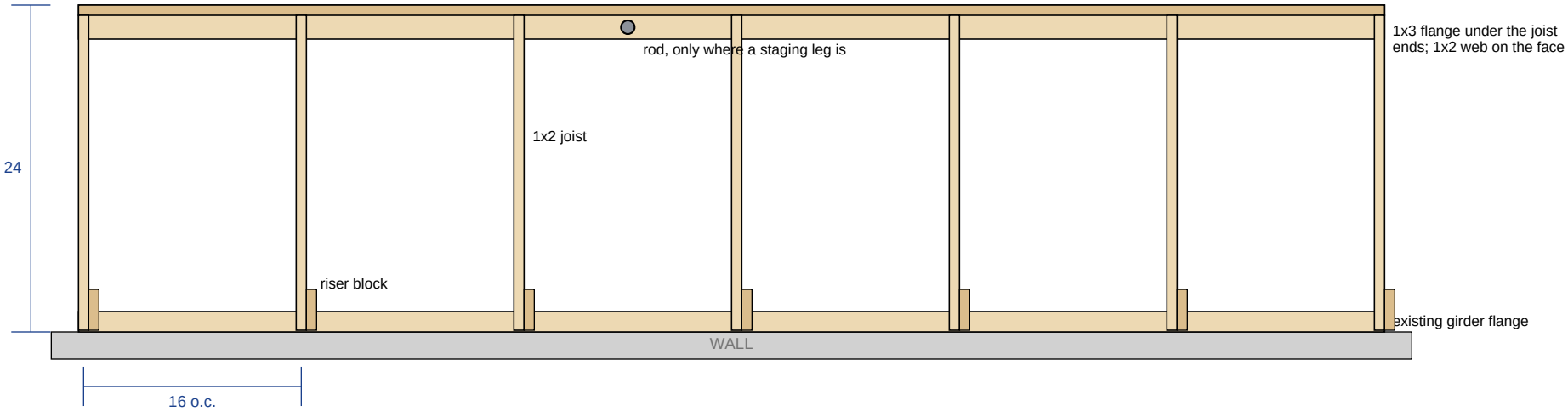


Riser block: cut from 1x3, grain running with the joist. Top flush with the joist top; the bottom sits on the flange. The wall end stays square; a 1 x 1 45 deg chamfer on the bottom front corner sits below the joist, so the glue face is the full 1-1/2 x 3 and the 2 in bottom bears on the whole 1-1/2 flange. Joist and block ends sit 1/8 off the wall. On the peninsula girders the block is centred on the flange, chamfer toward the aisle. Same side on every joist. Pattern on sheet 6.

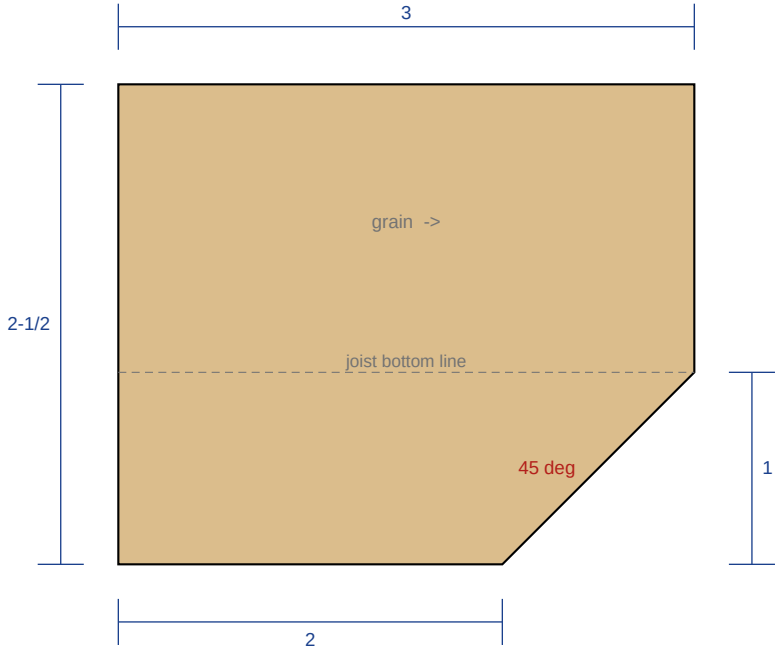
Plan - typical 8 ft bay, west wall, deck removed + riser block pattern

Looking down. Joists about 16" o.c.: riser blocks at the wall, butted to the front web.

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).



RISER BLOCK - FULL SIZE PATTERN (1:1)



square end goes to the wall

PER 8 FT BAY (24" deep, west wall)

1x2 joists	7 @ 23-1/8	wall gap 1/8 to the front web
1x3 riser blocks	7 @ 3	square at wall, 1" chamfer bottom front
1x2 front web	96	on edge, on the face
1x3 front flange	96	flat, under web + joists
3/4 ply	24 x 96	one rip per bay

The front girders go in as one 16 ft piece per run, no splice. Butt joints in the 3/4 ply land on a joist; the joist stations on sheet 1 already sit under every seam.

Scales: plan 1:12, pattern 1:1 - print at 100% and check the 3" dimension with a rule before tracing.

Assembly sequence, fasteners and lumber

Dimensions in inches. 0.00 = top of 3/4 ply subroadbed (0" grade).

1. Build both front girders on the bench: 1x3 flange flat, 1x2 web on edge on top - outer edge for the west girder, centred for the north - glue + screw up every 8". One 16 ft piece of each per run. Leave the west web out where the NW joist and the rim cross (H), and run it 7/8 past the flange's north end (F).

2. Mark the staging legs under each front edge. Pick two per run, keeping every gap under 60". Drill 7/16 holes through the flange there, between joist stations (detail D).

3. North girder: seat the west end on the west wall girder (E), the east end on the seat block at the wall-stub end stud (G). Prop at mid-span with the web top at -0.72.

4. West girder: seat the south end across the peninsula's north girder (E), butt the north end into the north girder and fit the gusset (F). Prop with the web top at -0.72.

5. Run the rods to the staging deck. Level both girders with the nuts, then lock with the opposing nuts.

6. Cut the riser blocks (pattern, sheet 6) and glue + screw one to each joist that lands on a -3.25 girder: the back wall end of every north and west joist, and both peninsula girders.

7. Set the joists at the sheet 1 stations, butted to the front web; screw up through the front flange into each joist and up through the back flange into each block. North: a 1x2 lookout in line with each joist. The rim and the NW joist run on past the west girder (H).

8. Fit the seam blocking (H). Build the closet boxes (north, south, then the short east box) and screw their wall sides into studs; then the SW closet strip: ledger, rail, joists.

9. Dry-lay the 3/4 ply; seams on joists or blocking. Glue + screw every 8" along each joist, keeping screws out from under the track centrelines (feeders). Fascia last.

FASTENERS

Girder flange to web	glue + #6 x 1-1/4 @ 8"
Flange up into joist	#6 x 1-1/4, pilot
Block to joist	glue + 2x #6 x 1-1/4
Back flange up into block	#6 x 1-1/2, pilot
Seat (E), through shim	glue + 2x #8 x 3
Gusset (F)	glue + 3x #6 x 1-1/4 per flange
Seat block (G)	2x #8 x 2-1/2, 1x #8 x 3
Rim, blocking ends	2x #6 x 2
Ply to joists	glue + #6 x 1-1/4 @ 8"
Rods	3/8 nut + fender washer each side, 7/16 hole

LUMBER (from the framing plan)

1x2, 16 ft	2	front girder webs
1x3, 16 ft	2	front girder flanges
1x2, 8 ft	13	30 joists, 11 lookouts, rim, blocking
1x3, 8 ft	4	39 riser blocks, closet boxes, SW strip
1x3 on hand	3 x 12 ft	+ 1 x 9 ft, cut first
1x2 / 1x3 net	117 / 95 ft	before offcuts
3/8-16 rod	2 x 72"	4 pieces ~15" + 2 under B1
Nuts / fender washers	24 / 24	
1/2 ply scrap	7-1/2 sq	gusset (F)
1/4 hardboard offcut	2 @ 2-1/2 sq	shims (E)

CLEARANCE OVER STAGING

Under joists	12.38 in
Under front girder	11.63 in
Under back girder web	7.10 in

Railhead = staging deck + 0.236 cork + 0.160 track.